Exercise 1: Map the Roman Empire

> Note
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> **Dataset:** Roman Empire Infrastructure

In this exercise you make two maps of the Roman Empire and a print layout with both maps. You practise loading raster and vector data, layer order, raster styling, categorized and graduated styles, labels, pie charts, map themes, coordinate systems and print layouts.

A) Map the empire and its landscape

You can do this after chapter 4.

1. Load all layers of the dataset. Put them in a logical order.
2. Show `elevation`: low land in green, mountains in brown or red. Use *Identify* to find the elevation of a city in the Alps.
3. Show the provinces with an outline only, so that the elevation stays visible.
4. Show the roads with a categorized style: major roads thicker than minor roads.
5. Show the cities with a categorized style on `type`: cities as small dots, legionary fortresses as squares. Label the cities with their `name`.
6. Save the current styles as a map theme.

B) Map the infrastructure per province

You can do this after chapter 4.

1. Restyle the provinces. Make sure you don't lose your previous style: first create a new style, then start configuring.
2. Colour the provinces with a graduated style on `area_km2`: the larger the province, the darker the colour. Label the provinces with their `name`.
3. Add a pie chart to every province with the number of monuments of each type (`n_theatres`, `n_amphitheatres`, `n_circuses`, `n_baths`, `n_temples`). The more monuments in total (`n_monuments`), the larger the pie chart.
4. Save the current styles as a map theme. Switch between the two map themes to check that both maps are kept.
5. Save the project in the folder of the dataset.

C) Prepare for finishing

You can do this after chapter 5.

1. Open your project of parts A and B again.
2. Check the CRS of every layer and of the project. Do all layers have the same CRS? Do they match the CRS of the project? What does this mean for your map?
3. Set the project to a projected CRS that suits the empire and compare the shapes of the provinces before and after.
4. Save the project again. Keep it: you need it in the next step.

D) Make a print layout

You can do this after chapter 6.

Open your project of parts A–C again. Make a finished map, as a PDF, with:

1. An overview map of the whole empire, showing the map you made in part B (provinces and pie charts). This is the largest map on the page. It shows a frame with the extent of the detail map.
2. A detail map that zooms in on the province of Belgica, showing the map you made in part A (elevation, roads and cities).
3. A title, and a legend, a scale bar and a north arrow that are linked to the overview map.